

Tools of the Trade: Linux and SQL

Module 1
Introduction to operating systems

Module 2
The Linux operating system

All about Linux

- Welcome to module 2
 - Video: 5:34
- Introduction to Linux
 - Video: 1:04
- File system and how to view the filesystems file
 - Video: 1:40
- Linux architecture
 - Video: 1:00
- Linux architecture explained
 - Reading: 1:00
- Adding components of the Linux OS
 - Reading: 1:00
- Test your knowledge: All about Linux
 - Practice assignment: 10:00

Linux distributions

- Linux Distributions
 - Video: 1:00
- RHEL/CentOS
 - Video: 1:00
- Which Linux Distributions
 - Reading: 1:00
- Managing packages for installing applications
 - Reading: 1:00
- Managing files and managing their roles
 - Reading: 1:00
- Link files and file sharing
 - Reading: 1:00
- Activating and installing a Linux distribution
 - Reading: 1:00
- Optional: Installing a Linux distribution
 - Reading: 1:00
- Managing local software in a Linux distribution
 - Reading: 1:00
- Test your knowledge: Linux distributions
 - Practice assignment: 10:00

The shell

- Introduction to the shell
 - Video: 1:00
- Different types of shells
 - Reading: 1:00
- Input and output to the shell
 - Video: 1:00
- Activating interactive input and output to the shell
 - Reading: 1:00
- Optional: Installing a Linux distribution
 - Reading: 1:00

Exemplar: Examine input and output in the shell

- Reading: 1:00

Test your knowledge: The shell

- Practice assignment: 1:00

Review: The Linux operating system

- Review
 - Video: 5:34
- Discovery: Terms from module 2
 - Reading: 1:00
- Module 2 challenge
 - Challenge assignment: 10:00

Module 3
Linux commands in the Bash shell

Module 4
Bash and SQL

Exemplar: Examine input and output in the shell

00:00 / 1:00

Activity overview

Previously, you discussed how the Bash shell helps you communicate with a computer's operating system. When you communicate with the shell, the computer in the shell can take input and return output as error messages.

In this activity, you'll use the echo command to examine how input is received and how output is returned to the shell. Next, you'll use the expr command to further explore input and output while performing some basic calculations in the shell.

The goal of this Bash foundation is understanding how you communicate with the Linux operating system through the shell. As a security analyst, you'll need to input commands into the shell and recognize when the shell returns other output or error messages.

Next, you'll explore the scenario!

This exemplar is a walkthrough of the previous Default Bash activity, including detailed instructions and solutions, but you can also attempt it yourself to complete the task and see your own progress in completing the task. You may also refer to this exemplar to prepare for the graded quiz in this module.

Scenario

As a security professional, it's important to understand the concept of communicating with your computer via the shell.

In this scenario, you have to input a specified string of text that you want the shell to return as output. Next, you also need to input a few mathematical calculations to the OS (operating system) can return the result.

Next, you will also find that you can use the **expr** command to generate some output to the shell. **Second**, you'll use the **expr** command to perform basic mathematical calculations. **Next**, you'll use the **expr** command to clear the Bash shell and **finally**, you'll have an opportunity to explore the **expr** and **expr** commands further.

Get ready to examine input and output in the Bash shell!

Task 1. Generate output with the echo command

The **echo** command in the Bash shell accepts a specified string of text. In this task, you'll use the **echo** command to generate output in the Bash shell.

1. Type **echo** into the shell and press **ENTER**

The command to complete this step

1 echo "hello"

The following string should be returned

1 hello

The command **echo** is the **input** to the shell, and **hello** is the **output** from the shell.

The command **echo** includes quotation marks around the string data. Type **echo "hello"** into the shell and press **ENTER**.

The command to complete this step

1 echo "hello"

The following string should be returned again

1 hello

Note: The output in the same buffer. The quotation marks are **optional** in this case, but they tell the shell to parse a series of characters together. This can be useful if you need to pass a string that contains certain characters that might be otherwise interpreted by the command.

2. Use the **echo** command to output your name to the shell.

Type **echo "name"** into the shell, replacing "name" with your own name, and press **ENTER**.

The command to complete this step

1 echo "your name"

The same you entered as the string should return as the output.

Task 2. Generate output with the expr command

In this task, you'll use the **expr** command to generate some output in the Bash shell. The **expr** command performs basic mathematical calculations and can be useful when you need to quickly perform a calculation. Imagine that the system has shown you that you have 10 shells, but only 1 required action. You need to calculate how many shells are left after you have 10 shells and you need to calculate the total number of shells that you have.

To do this, you need to subtract the number of shells that you need from the total number of shells.

1. Calculate the number of shells remaining using the **expr** command.

Type **expr 10 - 1** into the shell and press **ENTER**.

The command to complete this step

1 expr 10 - 1

The following result should be returned

1 9

Note: The **expr** command requires that all terms and operators in an expression are separated by spaces. For example, **expr 10 - 1** is not **expr 10-1**.

Now, you need to calculate the average number of shells that are expected over the course of a year. From the information you have, you know that an average of 100 shells are expected each month or so for this year.

So, you should be able to calculate the total number of shells expected in a year by multiplying **100** by **12**.

2. Type **expr 100 * 12** into the shell and press **ENTER**.

The command to complete this step

1 expr 100 * 12

The correct result should now be returned

1 1200

Task 3. Clear the Bash shell

In this task, you'll use the **clear** command to clear the Bash shell of all existing output. This allows you to start with the same as the top of the Bash shell window.

When you work in a shell environment, the screen can fill with previous input and output data. This can make it difficult to process what you're working on. Clearing the screen allows you to create a better view of the environment to allow you to focus on what is important at that point in time.

- Type **clear** into the shell and press **ENTER**.

The command to complete this step

1 clear

Note: All previous commands and output will be cleared, and the cursor position and cursor will return to the upper left of the shell window.

Optional task: Perform more calculations with the **expr** command

You can use the **expr** command to perform input and output further using **expr** and **expr** commands.

1. Generate a basic line-number output using the **expr** command.

(Remember the **echo** "hello" output you generated.)

The command to complete this step

1 echo "line number"

2. Perform at least one more calculation using the **expr** command.

The mathematical operations you can use with the **expr** command for **adding**, **subtracting**, **dividing**, and **multiplying** are **+**, **-**, **/**, and *****.

Note: The **expr** command performs integer mathematical calculations only, so you cannot use the decimal point or exponent notation. All results are rounded down to the nearest integer. Also, all terms and operators in an expression must be separated by spaces. For example, **expr 20 + 15**, or **expr 20*15**.

Conclusion

Great work!

This Bash foundation of experience is using basic Linux Bash shell commands to

- generate output with the **echo** command,
- generate output with the **expr** command, and
- clear the Bash shell with the **clear** command.

Understanding input and output is essential when communicating through the shell. It's important that you're comfortable with these basic concepts before you go on to work with additional commands.

Next in sequence

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Show me practice questions

Show me real-life examples

Ask me anything

1 out of 1 comments for this task. Check for mistakes and test if there are any. Your feedback will be used to improve with AI-powered assistance.

See next item

https://www.coursera.org/learn/linux-and-sql/supplement/vF4iE/exemplar-examine-input-and-output-in-the-shell

1/1